QUBO Formulation: VLSI Bisection Placement Problem

VLSI Bisection Placement Problem

Problem Statement. Given a set of N circuit modules and a set of weighted net connections E between them, the VLSI Bisection Placement Problem requires partitioning the modules into two placement regions, A and B, such that each region contains exactly $N/2$ modules. The objective is to minimize the total weight of nets that connect a module in region A to a module in region B.

Decision Variables. For each module $i \in \{0, 1, \ldots, N-1\}$, we introduce a binary variable $x_i \in \{0, 1\}$. The value $x_i = 0$ indicates that module i is assigned to region A, while $x_i = 1$ indicates assignment to region B.

QUBO Derivation. Step 1 — Write the original objective in general notation. A net $(i, j) \in E$ with weight w_{ij} crosses the bisection if and only if $x_i \neq x_j$. This condition is captured by the algebraic expression $x_i + x_j - 2x_i x_j$, which evaluates to 1 when $x_i \neq x_j$ and 0 otherwise. The minimization objective is therefore:

$$\min_{x \in \{0,1\}^N} \sum_{(i,j) \in E} w_{ij}(x_i + x_j - 2x_i x_j). \tag{1}$$

Step 2 — Write each constraint in its original algebraic form. The requirement that both regions contain exactly half of the modules is expressed as:

$$\sum_{i=0}^{N-1} x_i = \frac{N}{2}. \tag{2}$$

Step 3 — Convert the constraint to a penalty term. We enforce the constraint by adding a quadratic penalty term with weight $\lambda > 0$:

$$P(x) = \lambda \left(\sum_{i=0}^{N-1} x_i - \frac{N}{2} \right)^2. \tag{3}$$

Expanding the square and applying the idempotent property $x_i^2 = x_i$ for binary variables yields:

$$P(x) = \lambda \left(\sum_{i=0}^{N-1} x_i^2 - N \sum_{i=0}^{N-1} x_i + \frac{N^2}{4} + \sum_{i \neq j} x_i x_j \right) \tag{4}$$

$$= \lambda \left((1-N) \sum_{i=0}^{N-1} x_i + \sum_{i \neq j} x_i x_j + \frac{N^2}{4} \right). \tag{5}$$

Step 4 — Combine objective and penalty terms into the single QUBO Hamiltonian $H(x)$:

$$H(x) = \sum_{(i,j) \in E} w_{ij}(x_i + x_j - 2x_i x_j) + \lambda \left((1-N) \sum_{i=0}^{N-1} x_i + \sum_{i \neq j} x_i x_j + \frac{N^2}{4} \right). \tag{6}$$

Step 5 — Read off the Q matrix entries and offset c from H in general closed form. Collecting coefficients for each monomial $x_i x_j$ (with $i \leq j$) gives:

$$Q_{i,i} = \sum_{j:(i,j) \in E} w_{ij} + \lambda(1-N), \quad \forall i \in \{0, \ldots, N-1\}, \tag{7}$$

$$Q_{i,j} = -2w_{ij} + 2\lambda, \quad \forall i < j \text{ such that } (i,j) \in E, \tag{8}$$

$$Q_{i,j} = 2\lambda, \quad \forall i < j \text{ such that } (i,j) \notin E, \tag{9}$$

$$c = \lambda \frac{N^2}{4}. \tag{10}$$

Penalty Weight Justification. The penalty coefficient λ must be chosen sufficiently large to ensure that any violation of the balance constraint incurs a higher energy cost than the maximum possible reduction in the cut weight. Specifically, let $W_{\max} = \max_{(i,j) \in E} w_{ij}$ denote the maximum net weight. Choosing $\lambda > W_{\max}$ guarantees that the penalty for deviating from the required partition size strictly dominates any objective improvement achievable by an unbalanced assignment.

Correctness Argument. Minimizing $H(x)$ over $\{0,1\}^N$ recovers the optimal bisection because feasible solutions (those satisfying $\sum_i x_i = N/2$) incur zero penalty and their energy equals the cut weight. Infeasible solutions incur a penalty of at least λ. Since $\lambda > W_{\max}$, the energy of any infeasible solution exceeds the energy of the best feasible solution by a margin that cannot be offset by reducing the cut weight. Consequently, the global minimum of $H(x)$ must correspond to a feasible assignment that minimizes the total cut weight.

Illustrative Example. Consider a minimal instance with $N = 2$ modules and a single net connecting module 0 and module 1 with weight w_{01}. The constraint requires exactly one

module in region B, i.e., $x_0 + x_1 = 1$. Substituting $N = 2$ into the general formulas yields the concrete penalty expansion:

$$P(x) = \lambda\left((1-2)(x_0+x_1) + 2x_0x_1 + 1\right) = \lambda\left(1 - x_0 - x_1 + 2x_0x_1\right). \tag{11}$$

Combining this with the objective $w_{01}(x_0 + x_1 - 2x_0x_1)$ gives the full Hamiltonian:

$$H(x) = (w_{01}+\lambda)(x_0+x_1) - 2(w_{01}+\lambda)x_0x_1 + \lambda. \tag{12}$$

The corresponding QUBO parameters are:

$$Q_{0,0} = w_{01} + \lambda, \qquad Q_{1,1} = w_{01} + \lambda, \tag{13}$$

$$Q_{0,1} = -2(w_{01}+\lambda), \qquad c = \lambda. \tag{14}$$

Evaluating $H(x)$ over all four binary assignments reveals that the configurations $(x_0, x_1) = (0,1)$ and $(1,0)$ yield energy $w_{01} + \lambda - 2(w_{01}+\lambda) + \lambda = -w_{01} + \lambda$, while the balanced configurations $(0,0)$ and $(1,1)$ yield energy $w_{01} + \lambda$. Since $w_{01} > 0$, the minimum energy is $-w_{01} + \lambda$, correctly identifying the optimal bisection where the single net is cut, with minimum cut weight w_{01}.